Nubank Analytics Engineer Case

Introduction

You are in the next stage of the Nubank Analytics Engineer hiring process and now it is time to work in a hands-on project case.

The idea here is to show your knowledge based on previous experiences in fields that are relevant to this position applied to a very similar context to what happens at our company.

Data Architecture Overview

Here at Nubank, we have three different environments (Figure 1):

1. The **Production Environment** is where Nubank's services live. In this environment, services are responsible for getting customers' information and storing them in their respective databases.

2. Then, the datasets generated by the services are maintained in the **Data Warehouse Environment**, where Business Analysts, Analytics Engineers and any other Nubanker create new tables, improve data models and work to turn raw data into easily consumable tables for analysis.

3. Finally, these tables feed into the **Reporting Environment**, where it is possible to create all sorts of dashboards and visualizations for monitoring, decision-making, processes and so on.

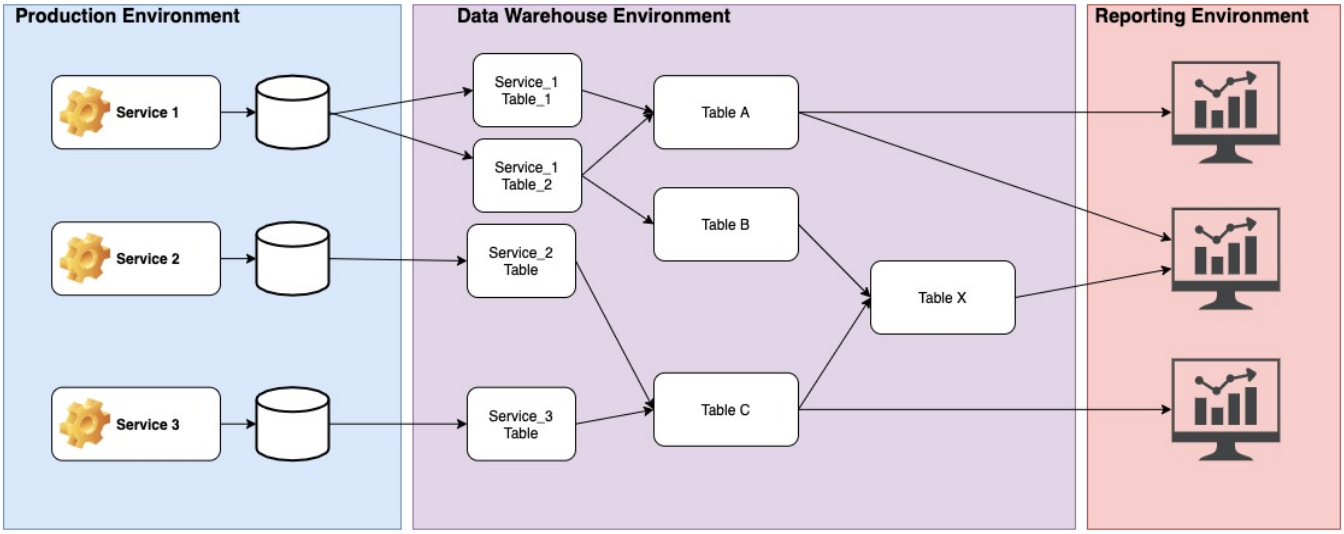

Figure 1 *- Nubank's environments: Production, Data Warehouse and Reporting.* A slice of the table structure from the Data Warehouse Environment is depicted in Figure 2 below.

Apart from time (d_time, d_year, d_month, d_week, d_weekday), location (city, state, country), accounts, and customers tables, three tables store the financial movements of the accounts:

- **transfer_ins:** non PIX transfers made to an account (money arriving)
- **transfer_outs:** non PIX transfers made from an account (money leaving)
- **pix_movements:** transfers that are either received by or sent from an account using PIX

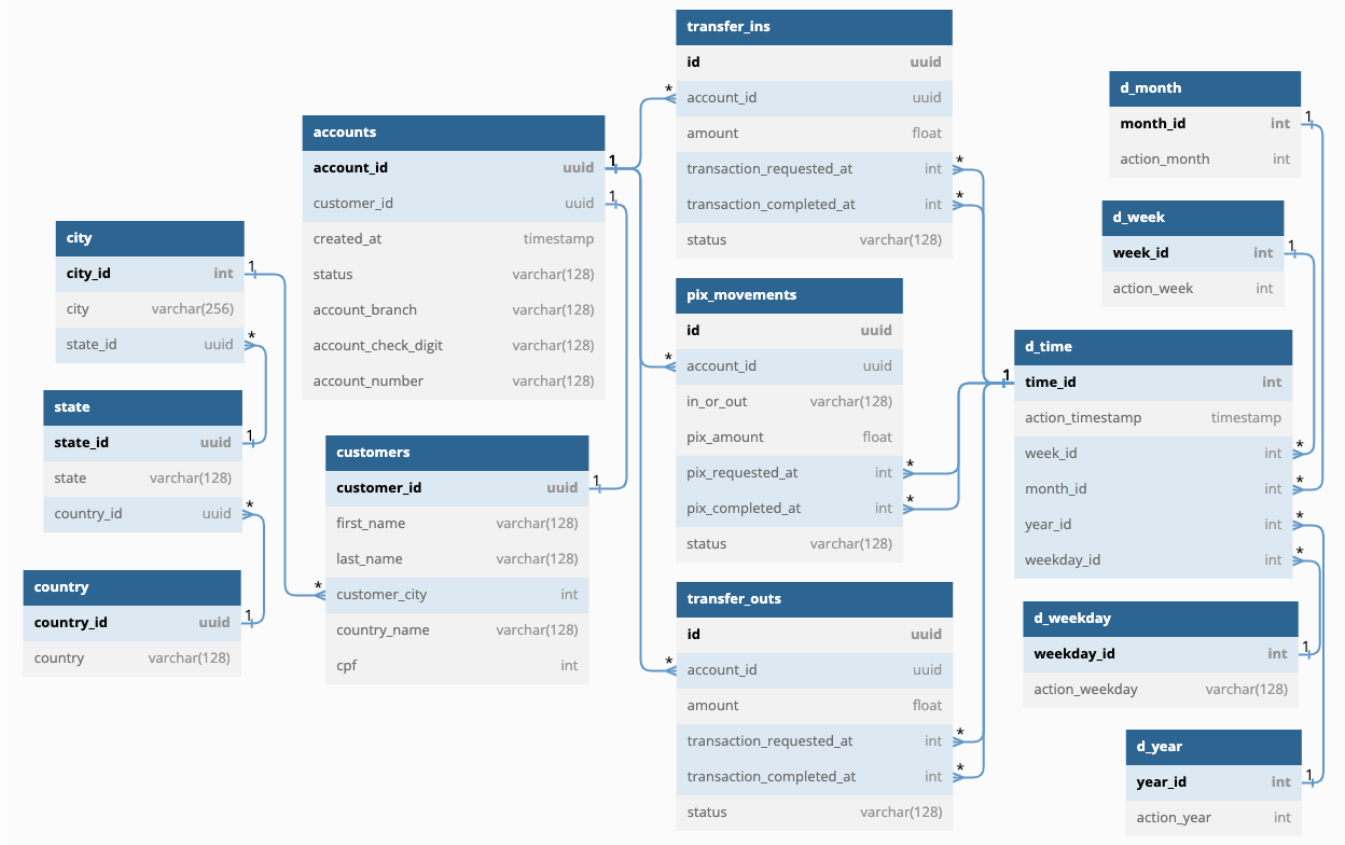

Figure 2 *- A slice of the table structure from the Data Warehouse Environment (diagram/table_diagram.png). You can check the code used to generate these diagrams using* ***dbdiagram.io*** *on file (diagram/table_diagram.txt).*

Problem Statement

Imagine that we would present the solution to either a **business audience** or a **technical audience** and people might access your solution in a moment that you are not available afterwards as well, so your solution should be **self-explanatory** as best as possible :)

You are asked to propose an improved data model for the table structure we presented in the Fig. 2. You may also consider that **Nubank is always evolving** with **new products** and it is also expanding to **new countries**, so our data warehouse model needs to accommodate all these incoming changes.

Keep in mind that the new products sometimes are not related to peer-to-peer transactions – for example: life insurance, lending, rewards and other products – and some of them might be available only in some countries. Knowing all of that, **which modifications** would you propose to the current data model and **why**?

Main Deliverables: The data model modification proposal having the **above introduction** in mind with:

- **visual representation** of the proposed changes (you can use **dbdiagram.io** if you like),
- and **trade-off analysis** of the proposed changes or **what was left the same**

Guidelines:

- **there are no right and wrong answers**, but we expect you to explain the rationale behind your decisions, the trade-offs involved
- Remember that **other analysts** will be using your proposed structure - your improvements should improves the lives of our analysts and they should be as clear as possible;

- **Feel free to change**, remove and/or add tables and fields to generate a better data model design.
- If you **decide to not change** some part of the data warehouse, we also expect a written explanation of why you decided to maintain it.

Here are some tips that might help you create your solution:

- First of all, **believe in yourself**! We're presenting you this case because we think you're capable of coming up with an interesting solution.
- When solving this case keep in mind that **Analytics Engineers** are the **bridge between the technical and business worlds**. This means that we're not only responsible for technical solutions and implementation details, but we should also consider business risks and impacts of our decisions.
- As Analytics Engineers, we care a lot about data governance. Also, having **clear documentation** is key for us to **communicate** with technical and non-technical stakeholders.
- We love to discuss our ideas and find the best solutions. So keep in mind that, **if your case gets accepted**, we'll **schedule a talk so we can debate your solution**. In this talk, we'll be going through your case and looking for the rationale behind it.
- Like every good documentation, we expect your **solution to be self-explanatory**, with **enough detail so readers can understand your rationale**. If you want to send additional files that would add value to your solution or use images, feel free to do so.

Suggestions for presentation

- The presentation must translate your thinking method to solve the case
- You will be required to explain your work
- Many questions will be made throughout the presentation, so **make sure to account time for discussions**
- The presentation should be easy to understand for non-tech people

Glossary

Account info (branch, number and check-digit)
In Brazil, a bank account can be uniquely identified by three numbers. Firstly the **branch** code that identifies in which bank branch the accounts were created. Secondly, the **account** number that identifies an account inside a branch. Finally, the **check-digit**, that exists only for error detection.

CPF
It is the Brazilian individual taxpayer registry identification.

PIX
This is the newest way to transfer money in Brazil. It's free. It is instantaneous, and one only needs to know the Pix-Key related to the account in order to make a transaction.

Non PIX transfers
Those are the traditional ways to move money from one bank account to another. In order to make this kind of transaction, one needs to inform the CPF, the branch code, the account-number and the check-digit of the account that will receive the money. The transaction usually takes several hours up to days to be confirmed, and most of the banks charge a fee in these transactions.